

Vision-Language Models for Embodied Reasoning

Abstract

Vision-language models have become central to embodied AI because they offer a plausible route from web-scale semantic knowledge to situated perception, task planning, and control. Yet “embodied reasoning” is not simply visual question answering with actions attached. It involves grounding natural-language goals in partial observations, maintaining memory over long horizons, inferring affordances and spatial relations, predicting counterfactual futures, and choosing actions that remain feasible under embodiment-specific constraints. Recent progress spans at least three partially overlapping families: modular planner-controller systems that use language models or vision-language models as high-level reasoners over tool APIs and skills; end-to-end vision-language-action policies that map multimodal histories directly to actions; and 3D or world-model-based systems that explicitly reason over geometry, map structure, or imagined futures. This survey synthesizes these strands under the lens of embodied reasoning rather than under narrower rubrics such as manipulation, navigation, or multimodal pretraining alone [1–4]. ¹

The survey argues that current progress rests on a productive but unresolved tension. On one side, internet-pretrained VLMs and MLLMs contribute rich semantic priors, open-vocabulary perception, and instruction following. On the other, embodied competence still depends on robotics-specific ingredients: action abstractions, temporal credit assignment, long-horizon memory, embodiment-aware controllers, simulation and teleoperation data, and evaluation protocols that expose failures in grounding, spatial reasoning, and safety. Benchmarks released in 2024–2026 make that tension particularly visible: strong models remain brittle on low-level manipulation, embodied spatial reasoning, free-exploration safety inspection, and navigation diagnostics, even when they appear capable in static multimodal QA. The field’s most promising direction is therefore not merely bigger VLMs, but tighter integration of structured perception, action-conditioned prediction, memory, and verification into unified embodied systems [25–47, 71–94, 106–125]. ²

Introduction

Embodied reasoning concerns how an agent transforms multimodal observations and language goals into temporally extended, physically grounded behavior. Unlike static vision-language inference, the embodied setting introduces partial observability, accumulated interaction history, embodiment-specific action spaces, distribution shift between training and deployment scenes, and safety-critical consequences of error. A robot or embodied simulator agent must not only recognize objects named in an instruction, but also infer latent task state, remember what has already been done, choose where to look, decide which action abstraction is appropriate, and revise plans when the world does not conform to textual priors. This richer notion of reasoning explains why progress has proceeded through multiple neighboring threads—vision-language pretraining, language-guided planning, robot imitation learning, world models, embodied navigation, and 3D scene understanding—rather than through a single model family [1–4, 15–24, 54–80]. ³

Historically, the field grew from at least two roots. The first was embodied vision-language understanding, especially navigation and household instruction following, where benchmarks such as Matterport3D, Room-to-Room, ALFRED, TEACH, BEHAVIOR, and Habitat emphasized grounding, memory, and long-horizon task structure [49–50, 54–67, 95–100]. The second was language-conditioned robot manipulation, where studies on affordance grounding, imitation over unstructured data, and scalable transformer policies culminated in RT-1, RT-2, Octo, OpenVLA, and related VLA systems [15, 25–47, 101–105]. What changed after 2022 was not only scale, but interface unification: the same pretrained visual and language backbones began to support planning, perception, and control within a shared tokenized architecture or within tightly coupled planner-controller loops [10, 13–23, 25–47]. ⁴

A crucial conceptual distinction is between semantic transfer and embodied transfer. Semantic transfer means that a model can use web-scale knowledge—for example, that mugs belong near coffee machines or that sponges are appropriate for spills. Embodied transfer means that the system can convert such knowledge into executable behavior under the robot’s sensors, kinematics, control frequency, and spatial context. RT-2 demonstrated that web knowledge can improve robotic control, OpenVLA showed that open-source VLAs can transfer across embodiments and tasks, and more recent reasoning-augmented VLAs have shown that intermediate deliberation or foresight can improve manipulation robustness. But benchmarks such as EmbodiedBench, EmbSpatial-Bench, BEAR, NaviTrace, and EPIC-Bench also show that high-level semantic competence does not yet imply robust embodied spatial grounding or low-level physical reasoning [26, 29–31, 39–47, 78–94]. ⁵

This survey is organized around methods rather than tasks alone. The central thesis is that embodied reasoning can be factored into several recurring design commitments: how visual-language priors are imported; whether planning is modular or end-to-end; whether the representation is 2D, 3D, or map-centric; how memory and world prediction are implemented; and how data, simulators, and benchmarks shape what is learned. The goal is not to flatten important differences between navigation and manipulation, but to reveal the emerging common substrate across them [1–4, 106–115]. ⁶

Methods

Pretrained visual-language priors as the substrate for embodied reasoning

Modern embodied models inherit much of their semantic competence from internet-scale visual-language pretraining. CLIP, ALIGN, Flamingo, BLIP-2, instruction-tuned VLMs such as LLaVA-style systems, and embodied multimodal LLMs such as PaLM-E provided the representation and interface primitives that later embodied systems adapted for control, planning, or 3D reasoning [5–14]. The core benefit of this inheritance is open-vocabulary perception: objects, relations, and instruction fragments can be interpreted even when exact categories or phrasings were not present in robot-only data. This is why embodied systems increasingly use frozen or lightly adapted visual backbones, language backbones, or both, and concentrate task-specific learning on action heads, adapters, prompting schemes, or small recurrent modules [10–14, 48–53]. ⁷

However, transferring general multimodal priors into embodied settings is nontrivial because pretraining objectives underemphasize control-relevant invariances such as viewpoint persistence, occlusion handling, contact state, and action-conditioned scene change. A notable line of work therefore augments web priors with robot-specific representation learning: R3M, VIP, masked visual pretraining for robot learning, sensorimotor pretraining, and CLIPort all showed that visual features improve when they are shaped by

interaction data, reward structure, or object-centric manipulation goals [43–52]. Even ostensibly general VLMs often require robot-centric alignment to become useful for control. The recurring lesson is that internet semantics and embodied affordances are complementary rather than interchangeable [43–53].

8

A related trend is “visual prompting” for action without full policy training. PIVOT and related mark-based or point-based prompting methods convert embodied decisions into iterative visual selection problems that large VLMs can solve zero-shot or near zero-shot [22]. These methods are important because they expose a middle ground between pure prompting and fully learned control: the model reasons semantically, but a structured external interface constrains output into candidate trajectories, waypoints, or interaction regions. In embodied reasoning terms, they externalize the latent search space that end-to-end policies must otherwise internalize [22, 53].

9

Modular planner-controller architectures

The first major family of embodied reasoning systems treats language models or VLMs primarily as high-level planners. SayCan grounds language in a library of robot skills by combining LM likelihood with affordance values [15]. Language Models as Zero-Shot Planners decomposes broad tasks into executable mid-level plans [16]. Inner Monologue adds iterative feedback from perception and execution to update plans during runtime [17]. Code as Policies lets an LLM synthesize policy code over perception outputs and control APIs, while ProgPrompt structures plans as constrained pseudo-programs over environment and action symbols [18–19]. Instruct2Act, VoxPoser, ThinkBot, and related systems extend the same principle to richer multimodal instructions, 3D value maps, or missing-step recovery in instruction following [20–21, 24].

10

These modular systems have several enduring strengths. They are interpretable because reasoning traces, code, or explicit subgoals can be inspected. They are data-efficient because the planner can inherit web-scale knowledge without retraining the controller. They also support heterogeneous embodiments because the reasoning layer can stay fixed while skills or APIs change. For long-horizon tasks, this separation remains practical: the planner reasons over semantics and task structure, while the controller handles fast closed-loop execution [15–24, 26–31].

11

Their weaknesses are equally instructive. Each interface boundary—between planner and skill library, between textual subgoal and visual state, between symbolic action and continuous controller—can create mismatch. A planner can propose semantically sensible but infeasible actions, or can over-trust textual priors when the scene contradicts them. Latency also grows when reasoning, verification, and execution happen in separate loops. Much recent work on embodied reasoning can be read as an attempt to retain the deliberative strengths of modularity while collapsing interface mismatch, either by fine-tuning shared backbones or by making reasoning differentiable and action-conditioned [17–24, 39–47, 116–121].

12

End-to-end vision-language-action policies

The second major family directly models behavior as a multimodal sequence prediction problem. RT-1 framed large-scale language-conditioned robot control with a transformer trained on real-robot trajectories [25]. RT-2 then fused robotic action prediction with VLM-style semantic transfer, showing that web knowledge can improve robotic behavior [26]. Open X-Embodiment generalized this recipe by aggregating robot data across institutions and embodiments, and Octo provided an open-source generalist policy

trained on hundreds of thousands of trajectories [27–28]. OpenVLA made the VLA paradigm broadly accessible and showed strong cross-embodiment performance with a 7B open model trained on 970k robot demonstrations [29]. ¹³

A key methodological question in this family is how actions are represented. Some models tokenize actions autoregressively, some predict action chunks, and others use diffusion or flow matching over continuous action trajectories. RT-H introduces action hierarchies using language [31]. Diffusion Policy, 3D Diffusion Policy, and 3D Diffuser Actor model multimodal action distributions and benefit from richer spatial representations [40–43]. Recent flow-based VLAs such as $\pi 0$ push toward dexterous, multi-platform control [39]. These design choices are not cosmetic: they shape latency, temporal coherence, fine motor control, and the extent to which reasoning can be inserted before action decoding [31, 39–43]. ¹⁴

Another important question is how much of the backbone should be adapted. OpenVLA-OFT showed that adaptation details—parallel decoding, action chunking, continuous action representations, and objective choice—substantially affect both inference speed and success rate [30]. Other studies have revisited whether generic VLM encoders are sufficient as-is or whether control-specific supervision is needed to produce robust policies. This line of work suggests that the main bottleneck is no longer only pretraining scale; it is also how efficiently semantics are translated into fast, stable motor interfaces [29–30, 35–47]. ¹⁵

The most recent VLAs increasingly reincorporate explicit reasoning. Embodied chain-of-thought methods, CoT-VLA, and test-time sampling or value-guided methods introduce intermediate visual or textual deliberation before action generation [44–47]. Conceptually, these systems blur the old opposition between modular planning and end-to-end control. They still optimize a policy backbone, but they expose or train internal foresight, subgoal prediction, or verifier-guided search. In that sense, they represent the clearest current embodiment of “reasoning inside the policy” rather than “reasoning around the policy” [44–47]. ¹⁶

Navigation and spatial reasoning

Embodied navigation was the earliest sustained testbed for vision-language grounding, and it remains the clearest lens on spatial reasoning. The sequence from Matterport3D and Room-to-Room through Speaker-Follower, Self-Monitoring, PREVALENT, VLN-BERT, HAMT, DUET, BEVBERT, and ETPNav traces a methodological arc from instruction grounding and imitation learning to large-scale pretraining, history-aware transformers, and map-based planning [54–64]. Importantly, these papers established that memory, topological structure, and pretraining all matter, but in different ways: local grounding errors, exploration failures, and accumulation of orientation drift are distinct pathologies requiring different architectural remedies [54–64]. ¹⁷

A separate but connected line concerns task variation and richer supervision. RxR broadened linguistic and cultural diversity for navigation instructions [65]. CVDN added dialogic guidance [66]. Touchdown moved VLN into urban environments with harder spatial language [67]. Continuous-environment variants and object-goal navigation benchmarks revealed the gap between graph-structured training assumptions and real embodied motion [68–70]. These shifts matter for embodied reasoning because they expose when a model’s performance depends on benchmark regularities rather than on genuine spatial understanding [65–70]. ¹⁸

Since 2023, foundation-model navigation has diversified. NaviLLM casts heterogeneous navigation tasks into a unified schema-based generation format [71]. NavGPT and NaVid explicitly use LLM/VLM reasoning

or video-centric VLMs for next-step planning [72–73]. ImagineNav converts planning into VLM selection over imagined future views [74]. VLMnav tests whether a VLM can act as a direct end-to-end navigation policy with little or no navigation-specific training [75]. Nav-R1 brings chain-of-thought-style reasoning and reinforcement learning into embodied navigation, while MapNav and WorldMAP emphasize structured semantic memory and world-model-generated supervision [76–78]. Together these methods reveal a field-wide shift: navigation is no longer only an instruction-following problem, but increasingly a testbed for controllable embodied reasoning with memory, imagination, and verification [71–78]. ¹⁹

Recent evaluation work underscores how incomplete that shift remains. NaviTrace evaluates trace prediction in real-world images and finds persistent weaknesses in goal localization and spatial grounding [79]. VLN-MME reports that adding chain-of-thought or self-reflection to general MLLMs can actually hurt navigation, implying that generic reasoning traces do not automatically align with embodied context [80]. These results are a useful corrective: in embodied navigation, better-looking verbal reasoning is not equivalent to better situated control [79–80]. ²⁰

Three-dimensional grounding and embodied question answering

If navigation stresses sequential spatial control, 3D embodied reasoning stresses viewpoint, geometry, perspective, and scene memory. ScanQA and SQA3D illustrate the challenge sharply. ScanQA emphasizes open-ended question answering over real 3D scans; SQA3D requires a model to localize a described situation in the scene and answer from that perspective, with strong human-model gaps [81–82]. These tasks are especially relevant because many downstream embodied tasks reduce to latent 3D queries: Where is the object relative to me? Which receptacle is accessible? What changed after the previous action? [81–82]. ²¹

Models such as 3D-LLM, LEO, ShapeLLM, and Uni3D-LLM bring 3D point clouds, scene graphs, or geometric features into language-model interfaces [83–86]. Their importance lies less in raw benchmark scores than in changing what the reasoning substrate can represent. Rather than inferring geometry indirectly from 2D images alone, these models can align language with explicit spatial structures, support 3D grounding or captioning, and connect embodied interaction to object- and scene-level geometry [83–86]. LEO is especially notable for explicitly unifying perception, grounding, reasoning, planning, and acting in a 3D world interface [84]. ²²

3D-VLA extends this trend from reasoning to action. Instead of treating control as a direct 2D observation-to-action mapping, it uses a 3D vision-language-action generative world model that predicts goal images and point clouds, thereby connecting action choice to imagined future states [87]. This is conceptually significant because it reframes embodied reasoning as a joint problem of state abstraction and counterfactual generation. OpenEQA likewise shows that foundation-model-era embodied QA increasingly depends on exploration, not merely passive scene interpretation [88]. ²³

Data, simulators, and benchmarks

Embodied reasoning is unusually sensitive to data interface and evaluation protocol. Robot policy learning depends on expensive demonstration data, while reasoning-oriented benchmarks depend on language and scene curation choices that can bias results toward linguistic priors. Open X-Embodiment, Bridge Data, RoboNet, CALVIN, LIBERO, and RLBench each occupy a distinct niche in this ecosystem: cross-embodiment transfer, domain diversity, multi-robot reuse, long-horizon language-conditioned manipulation, lifelong

transfer, and broad multi-task simulation, respectively [27, 101–105]. Their coexistence reflects the fact that there is still no single canonical training distribution for general embodied reasoning [27, 101–105]. ²⁴

Simulators and household task suites play an equally important role. ALFRED, TEACH, BEHAVIOR, BEHAVIOR-1K, Habitat 2.0, and Habitat 3.0 each increase realism along different axes: action compositionality, dialogue, everyday household breadth, human-centered task coverage, physics-enabled rearrangement, and human-robot collaboration [95–100]. These benchmarks matter because they progressively remove shortcuts available in purely symbolic or short-horizon settings. At the same time, they raise the difficulty of fair comparison: a method that excels in pointwise semantic grounding may still fail when exposed to dialogue repair, clutter, or prolonged interaction [95–100]. ²⁵

The newest benchmarks focus less on broad capability and more on diagnostic failure analysis. ECBench targets embodied cognition in egocentric video [89]. EmbSpatial-Bench isolates embodied spatial understanding and shows clear deficits even in powerful LVLMs [90]. A 2025 study on embodied knowledge reports that visual grounding does not automatically yield stronger embodied knowledge understanding than text-only models [91]. EmbodiedBench reveals that leading MLLM agents remain weak on low-level manipulation, with GPT-4o achieving only 28.9% average score across its benchmark [92]. BEAR and EPIC-Bench expose persistent limits in atomic embodied capabilities and fine-grained visual grounding, respectively [93–94]. These benchmarks collectively shift the field from leaderboard chasing toward capability decomposition [89–94]. ²⁶

World models, memory, and long-horizon deliberation

Embodied reasoning becomes substantially harder once tasks extend beyond the immediate observation window. Sequence-modeling approaches such as Decision Transformer and Trajectory Transformer established a useful abstraction by treating control as conditional sequence modeling [106–107]. World-model methods such as Dreamer, DreamerV2, DreamerV3, and DayDreamer showed that learning predictive latent dynamics can improve efficiency and support “planning in imagination,” including on physical robots [108–111]. These works did not begin as vision-language systems, but they supplied key ideas now being absorbed into VLAs: latent rollouts, short-horizon prediction, and action-conditioned futures [106–111]. ²⁷

The recent convergence between VLMs and world models is one of the clearest signs that embodied reasoning is moving beyond reactive control. Learning Interactive Real-World Simulators demonstrated that diverse data sources can be orchestrated into interactive generative environments that support both high-level VLM policies and lower-level control [112]. Language Models Meet World Models argues in the opposite direction that embodied experience can enhance language models [113]. WorldMAP shows that world-model-generated futures can be converted into structured supervision for navigation [78]. The World Action Models survey formalizes this growing paradigm as the joint modeling of actions and future states rather than actions alone [114]. ²⁸

Memory is the other half of long-horizon reasoning. MEM explicitly separates short-horizon visual memory from longer-horizon textual memory for VLA control, enabling tasks lasting up to many minutes [115]. In practice, this type of multi-timescale memory addresses a recurring embodied bottleneck: recent frames are needed for fine motor correction, but symbolic summaries are often better for remembering abstract task progress. The broader literature increasingly treats memory not as a single recurrent hidden state, but

as a structured design space involving keyframe selection, semantic compression, map updates, retrieval, and user-specific history [71, 77–80, 115]. ²⁹

Challenges and Prospects

The most immediate challenge is perceptual grounding. Benchmarks released since 2024 consistently show that strong multimodal models remain brittle on embodied spatial relations, affordances, part-whole reasoning, trace prediction, and mask-grounded localization [89–94]. This implies that the main bottleneck is not generic language understanding, but the conversion of language into action-relevant visual structure. Future systems will likely need richer intermediate representations—object-centric state, affordance fields, spatial memories, and uncertainty-aware detections—rather than relying on free-form textual outputs alone [22, 53, 87, 94]. ³⁰

A second challenge is balancing deliberation with latency. Planner-controller stacks remain attractive for long-horizon reasoning, but they often incur interface mismatch and runtime overhead. End-to-end VLAs offer speed and unified training, but without explicit reasoning they can overfit to reactive mappings. The strongest recent results suggest that the productive frontier is hybrid: internalized foresight, action-conditioned verification, chunked action generation, and hierarchical memory can add reasoning without destroying control throughput [30–31, 39–47, 112–115]. ³¹

A third challenge is data. Cross-embodiment aggregation has been transformative, but large robotic datasets remain narrow relative to internet-scale vision-language corpora, and they undersample failures, hazards, and rare long-horizon interactions [27, 29, 54, 101–105]. Future progress will probably require more than just more teleoperation. The likely ingredients are structured simulation, human demonstration interfaces, egocentric videos, action-conditioned world-model rollouts, and synthetic supervision generated through 3D or video priors [36–39, 78, 87, 112–114]. ³²

A fourth challenge is evaluation. Contemporary embodied benchmarks increasingly demonstrate that broad averages can conceal category-specific deficiencies. A model can excel at high-level semantic planning while failing at low-level manipulation, or can appear safe only because it over-refuses. Diagnostic evaluation therefore needs to separate perception, memory, planning, spatial reasoning, action grounding, calibration, and abstention. The move from narrow navigation/manipulation leaderboards toward EmbodiedBench, ECBench, BEAR, EPIC-Bench, SafeAgentBench, EARBench, HomeSafeBench, HEAL, and EgoSafetyBench is a positive development because it begins to expose those factors separately [89–94, 116–121]. ³³

Safety and hallucination remain major open problems. SafeAgentBench reports that state-of-the-art embodied LLM agents reject only a small fraction of hazardous tasks, EARBench finds very high task risk rates across models, BadRobot shows that embodied agents can be jailbroken into harmful behavior, HomeSafeBench reveals severe weakness in hazard inspection, and HEAL demonstrates that scene-task inconsistencies can induce large hallucination rates [116–120]. These findings suggest that safety in embodied reasoning cannot be reduced to standard conversational alignment. It requires grounded infeasibility detection, runtime monitoring, action-level verification, and explicit uncertainty handling when perception, language, and task priors conflict [116–121]. ³⁴

The long-term prospect is a generalist embodied reasoner that unifies semantics, geometry, memory, foresight, and embodiment-aware control. Recent work on GR00T N1, Xiaomi-Robotics-0, Vesta, and SpatialCoT indicates several components of such a system: broad robot-policy pretraining, real-time

inference, unified navigation-manipulation-localization interfaces, and stronger spatial alignment for reasoning [122–125]. But these should be interpreted as early systems rather than as solved endpoints. The central research opportunity is to build models that can say not only “what should be done,” but also “what is physically possible here, now, under this embodiment, and with what confidence.” That is the essence of vision-language modeling for embodied reasoning. ³⁵

Conclusion

Vision-language models have changed embodied AI most profoundly not by replacing robotics with generic multimodal intelligence, but by forcing a new synthesis between them. Semantics from web-scale pretraining, structured interaction from robotics, geometric reasoning from 3D modeling, and imagination from world models are now converging into architectures that increasingly treat perception, reasoning, and action as a single coupled system [1–4, 25–47, 83–87, 112–115]. ³⁶

The survey’s main conclusion is therefore twofold. First, embodied reasoning is emerging as a unifying analytical lens across navigation, manipulation, agent planning, embodied QA, and safety evaluation. Second, the decisive advances ahead are likely to come from models that combine rich multimodal priors with explicit state structure, long-horizon memory, action-conditioned foresight, and fine-grained verification. The field has already moved beyond the question of whether VLMs can be used for embodied agents. The harder and more consequential question is how to turn them into reliable embodied reasoners.

37

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